Exercise 4: Worked solutions

Mathematics for Engineers I · Module 62765 · Exercise Sheet 4

From Fundamentals to AI Apocalypse · personal study guide

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Exercise 4.1: Mathematical induction

Induction starts from the first valid index and shows that each true case forces the next.

Connected topic: Induction (§2.5).

4.1a: Prove the finite geometric sum

Try first: Check n=0, then multiply the proposed sum by x and subtract, keeping $x \neq 1$ visible.

Answers: see "Exercise 4.1: Mathematical induction" under Answers at the end.

4.1b: Prove the reciprocal-square-root bound

Try first: Try induction from n=2: compare the new term with the change in $\sqrt{n}$, or use a direct termwise bound.

Answers: see "Exercise 4.1: Mathematical induction" under Answers at the end.

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Exercise 4.2: Maps

Composition applies f first and g second. For finite-set counts, choose an image for each input while respecting the required property.

Connected topic: Injectivity (§3.2) and finite counting (§2.4); Map types (§3.2) and counting and binomial coefficients (§2.4).

4.2a: Is a composition of injections injective?

Try first: Assume two composite outputs are equal and unwind the definition of injectivity in reverse order.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

4.2b(i): Count all maps M→N

Try first: Each of the m elements of M independently chooses one of n target elements.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

4.2b(ii): Count bijections M→N

Try first: A bijection requires equal cardinalities; when they match, count the possible orderings.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

4.2b(iii): Count injective maps M→N

Try first: An injection assigns distinct targets to inputs, so there are fewer choices at each step.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

4.2b(iv): Count surjective maps M→N

Try first: Exclude maps whose image misses at least one target, correcting for maps that miss several targets.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

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Exercise 4.3: Elementary functions

Addition formulas for sine and cosine imply the circle identity and double-angle relation.

Connected topic: Elementary functions and inverse functions (§3.5).

4.3a: Prove the circle equation

Try first: Expand $\cos(x-x)$ using the cosine addition theorem.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

4.3b: Prove the cosine double-angle identity

Try first: Use $\cos(2x)=\cos(x+x)$, then replace $\cos^2 x$ by $1-\sin^2 x$.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

4.3c: Evaluate cos(arcsin x)

Try first: Let $\theta=\arcsin x$, so $\sin\theta=x$ and θ lies in the principal interval where cosine is non-negative.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

4.3d: Find cos(π/5) exactly

Try first: Use $\pi/5=2(\pi/10)$ in the double-angle identity, then select the positive root because $\pi/5$ is in the first quadrant.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

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Self-Study 4.4: Maps

Vertex form reveals the range and the restriction needed for a quadratic to become one-to-one.

Connected topic: Injectivity and inverses (§3.2) and elementary functions (§3.5); Composition and domains (§3.4).

4.4a: Classify f, g, and h; modify f and g

Try first: Complete the square for f, solve g for its input, and inspect the coordinate relation defining h.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4.4b(i): Form f∘e

Try first: Since $e(x)=e^x$ is always positive, substitute it into f and use the range of the downward parabola.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4.4b(ii): Form g∘f

Try first: The inner output must satisfy $f(x)\geq 0$ because g accepts only non-negative inputs.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4.4b(iii): Form g∘e

Try first: Check the exponential output against g's non-negative domain.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4.4b(iv): Form g∘g

Try first: The first g maps $[0,\infty)$ into $(0,1]$, which is contained in the domain $[0,\infty)$ of the second g.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

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Self-Study 4.5: Maps

Use the definitions directly: equality of composite values, existence of preimages, and the order of quantifiers.

Connected topic: Injectivity and surjectivity (§3.2).

4.5i: If g∘f is injective, show f is injective

Try first: Assume f sends two inputs to the same value, then apply g.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

4.5ii: If g∘f is surjective, show g is surjective

Try first: Take an arbitrary element of Z and use surjectivity of the composite to produce a preimage.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

4.5iii: If f and g are surjective, show g∘f is surjective

Try first: For each target z, first choose a g-preimage y, then choose an f-preimage x of y.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

4.5iv: Give an injective composite with non-injective g

Try first: Make the domain of f a singleton, while g identifies two different middle elements.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

Self-Study 4.6: Elementary functions

Factor the quartic as a quadratic in x^2. For the parameterized quadratic, use the discriminant and include degenerate cases.

Connected topic: Polynomials and elementary functions (§3.5); Elementary functions (§3.5).

4.6a: Find and factor the roots

Try first: Set $u=x^2$ for the quartic, then split the parameter cases for g_a.

Answers: see "Self-Study 4.6: Elementary functions" under Answers at the end.

4.6b: Show cos(3x) is not generally 3cos x

Try first: A single convenient counterexample disproves a universal equality; then derive the general expression.

Answers: see "Self-Study 4.6: Elementary functions" under Answers at the end.

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Self-Study 4.7: Strictly monotone functions

Strict monotonicity forbids equal outputs at distinct inputs; injectivity alone does not require order preservation.

Connected topic: Injectivity and inverse functions (§3.2).

4.7a: Increasing implies injective

Try first: For two distinct inputs, order them and apply the strict inequality.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

4.7b: Decreasing implies injective

Try first: Repeat the order argument with inequalities reversed.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

4.7c: Is every injection strictly monotone?

Try first: Find an injective rearrangement of the real line that reverses order for one pair.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

4.7d: Conclude that exp has an inverse

Try first: Combine strict increase with the known range of the exponential.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

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Self-Study 4.8: Tangent addition theorems

Divide the sine and cosine addition identities only when the stated cosines are nonzero.

Connected topic: Trigonometric functions (§3.5).

4.8a: Prove the tangent addition and subtraction formulas

Try first: Write $\tan(x\pm y)=\sin(x\pm y)/\cos(x\pm y)$, then divide numerator and denominator by $\cos x \cos y$.

Answers: see "Self-Study 4.8: Tangent addition theorems" under Answers at the end.

4.8b: Derive the tangent double-angle formula

Try first: Set $y=x$ in the addition formula, observing the stated conditions keep the tangent defined.

Answers: see "Self-Study 4.8: Tangent addition theorems" under Answers at the end.

4.8c: Compute $\tan(\pi/12)$ exactly

Try first: Apply the subtraction formula to $\pi/3-\pi/4$ and simplify by rationalizing.

Answers: see "Self-Study 4.8: Tangent addition theorems" under Answers at the end.

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Challenge 4.A: Surjective maps

Let s(k) count maps onto a fixed k-element set. Partition all maps according to their image, then invert the resulting triangular relation.

Connected topic: Map types (§3.2) and counting and binomial coefficients (§2.4); Finite sums and binomial coefficients (§2.4); Binomial coefficients (§2.4); Binomial coefficients and finite sums (§2.4); Induction (§2.5) and map types (§3.2).

4.A setup: Interpret s(k) and the target

Try first: Relate surjectivity to the size of the image, and identify s(n) with the desired count.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 1: Derive the recurrence for s(k)

Try first: Partition the k^m maps by their exact image size j and solve for the j=k term.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 2: Check the proposed formula for k=1, 2, and 3

Try first: Use the recurrence to calculate the first values and compare them with the binomial sum.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 3a: Reverse the order of a triangular sum

Try first: List the index region rather than manipulating terms.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 3b: Prove the binomial coefficient identity

Try first: Write both sides using factorial definitions.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 3c: Establish the convolution identity

Try first: After changing order, the coefficient of each i contains a complete alternating binomial sum.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 4: Prove s(k)=g(k) by induction

Try first: Use k=1 as the base, then subtract the same lower-index terms from the recurrence and convolution identity.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

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Answers

Exercise 4.1: Mathematical induction

4.1a · Prove the finite geometric sum

1. Let $S_n=\Sigma_{k=0}{}^n x^k$. The claim is $S_n=(1-x^{n+1})/(1-x)$, for $x\neq 1$.
2. At $n=0$, $S_0=1$ and $(1-x)/(1-x)=1$.
3. Assume the formula for n. Add the next term: $S_{n+1}=S_n+x^{n+1}=(1-x^{n+1})/(1-x)+x^{n+1}$.
4. Put the terms over the same denominator: $[1-x^{n+1}+x^{n+1}(1-x)]/(1-x)=(1-x^{n+2})/(1-x)$.
5. That is the claimed formula with n+1. The denominator is nonzero because $x\neq 1$; the base and step prove the result for all $n\geq 0$.
6. $$\sum_{k=0}^{n} x^k \;=\; (1-x^{n+1})/(1-x)$$

Exercise 4.1: Mathematical induction

4.1b · Prove the reciprocal-square-root bound

1. For $1\leq k\leq n$, $\sqrt{k}\leq\sqrt{n}$, so $1/\sqrt{k}\geq 1/\sqrt{n}$.
2. Summing these n inequalities gives $\Sigma_{k=1}{}^n 1/\sqrt{k}\geq n/\sqrt{n}=\sqrt{n}$.
3. For $n\geq 2$ at least one inequality is strict (for example $k=1 k=1^n 1/\sqrt{k}>\sqrt{n}$.
4. This proves the stated claim directly, and also explains why the inequality is strict.
5. $$\sum_{k=1}^{n} 1/\sqrt{k} \;>\; \sqrt{n}$$

Exercise 4.2: Maps

4.2a · Is a composition of injections injective?

1. Suppose $(g\circ f)(x_1)=(g\circ f)(x_2)$. This says $g(f(x_1))=g(f(x_2))$.
2. Since g is injective, $f(x_1)=f(x_2)$. Since f is injective, $x_1=x_2$.
3. Thus $g\circ f$ is injective. The codomains need not be equal: the key is that each map is injective on the outputs that occur.
4. $$(g\circ f)(x_1) = (g\circ f)(x_2) \;\Rightarrow\; x_1 = x_2$$

Exercise 4.2: Maps

4.2b(i) · Count all maps M→N

1. For each input x∈M, a map may assign any of the n elements of N.
2. There are n choices for each of m independent inputs, so the product rule gives $n \cdot n \cdots n = n^m$ maps.
3. If n=0 and m>0, this gives 0 maps, as expected; for m=0 the unique empty map is counted by $0^0=1$ under the combinatorial convention.
4. $$\#(M \to N) = n^m$$

Exercise 4.2: Maps

4.2b(ii) · Count bijections M→N

1. If m≠n, no bijection exists, so the count is 0.
2. If m=n, list the m inputs in any fixed order. The first image has m choices, the next m−1, and so on to 1.
3. The product is m!, including 0!=1 for the empty sets.
4. $$\#\{\text{bijective maps } M \to N\} = m! \text{ when } m = n; \text{ otherwise } 0$$

Exercise 4.2: Maps

4.2b(iii) · Count injective maps M→N

1. If m>n, the pigeonhole principle makes an injection impossible; the count is 0.
2. For m≤n, order the inputs. Choose distinct images in sequence: n choices, then n−1, down to n−m+1.
3. Therefore the count is $n(n-1)\ldots(n-m+1) = n!/(n-m)!$. For m=0 this is 1.
4. $$\#\{\text{injective maps } M \to N\} = n!/(n-m)! \text{ when } m \leq n; \text{ otherwise } 0$$

Exercise 4.2: Maps

4.2b(iv) · Count surjective maps M→N

1. Start with all n^m maps. For a chosen set of j target elements forbidden from the image, the remaining n−j targets give $(n-j)^m$ maps.
2. There are C(n,j) ways to choose those j omitted targets. Inclusion–exclusion alternates the correction by the number omitted.
3. The number of onto maps is $\Sigma_{j=0}{}^{n}(-1)^j\ C(n,j)(n-j)^m$. This includes the empty-set cases when interpreted with the usual combinatorial convention.
4. $$\#\{\text{onto maps } M \to N\} = \sum_{j=0}^{n} (-1)^j\, C(n,j)(n-j)^m$$

Exercise 4.3: Elementary functions

4.3a · Prove the circle equation

1. The cosine difference formula is cos(a−b)=cos a cos b+sin a sin b.
2. Set a=b=x. The left side is cos 0=1, and the right side is $\cos^2 x+\sin^2 x$.
3. Hence $\cos^2 x+\sin^2 x=1$ for every real x.
4. $$\cos^2 x + \sin^2 x = 1$$

Exercise 4.3: Elementary functions

4.3b · Prove the cosine double-angle identity

1. The addition formula gives $\cos(x+x)=\cos^2 x-\sin^2 x$.
2. Using $\cos^2 x=1-\sin^2 x$ from the circle equation, $\cos(2x)=1-\sin^2 x-\sin^2 x=1-2\sin^2 x$.
3. $$\cos(2x) = 1 - 2\sin^2 x$$

Exercise 4.3: Elementary functions

4.3c · Evaluate cos(arcsin x)

1. By definition, $\theta=\arcsin x$ means $\sin\theta=x$, with $\theta\in[-\pi/2,\pi/2]$.
2. The circle identity gives $\cos^2\theta=1-\sin^2\theta=1-x^2$. On this interval $\cos\theta\geq 0$, so take the non-negative square root.
3. Therefore $\cos(\arcsin x)=\sqrt{(1-x^2)}$, valid for $-1\leq x\leq 1$.
4. $$\cos(\arcsin x) = \sqrt{(1-x^2)},\ -1 \leq x \leq 1$$

Exercise 4.3: Elementary functions

4.3d · Find cos(π/5) exactly

1. The given value is $\sin(\pi/10)=(\sqrt{5}-1)/4$. Its square is $(3-\sqrt{5})/8$.
2. Apply $\cos(2x)=1-2\sin^2 x$ with $x=\pi/10$: $\cos(\pi/5)=1-2(3-\sqrt{5})/8=(1+\sqrt{5})/4$.
3. The result is positive, as expected for $\pi/5\in(0,\pi/2)$.
4. $$\cos(\pi/5) = (1+\sqrt{5})/4$$

Self-Study 4.4: Maps

4.4a · Classify f, g, and h; modify f and g

1. For $f(x)=-x^2+6x-5$, complete the square: $f(x)=-(x-3)^2+4$. Its range is $(-\infty,4]$; it is not injective on $\mathbb{R}$ because points equally spaced about 3 have equal values.
2. Restrict f to $[3,\infty)$ and use codomain $(-\infty,4]$. It is then strictly decreasing and onto that codomain. Solving $y=4-(x-3)^2$ on this branch gives $f^{-1}(y)=3+\sqrt{(4-y)}$, $y\leq 4$.
3. For $g(x)=1/(1+x)$, $x\geq 0$, the values are $(0,1]$. It is strictly decreasing and injective, but not onto $\mathbb{R}$. With codomain $(0,1]$, it is bijective; solving $y=1/(1+x)$ gives $g^{-1}(y)=1/y-1$.
4. For $h(x,y)=(x,x+y,y)$, equality of outputs gives equality of the first and third coordinates, hence x and y agree: h is injective. Its image is the plane $\{(a,b,c):b=a+c\}$, so it is not onto $\mathbb{R}^3$ and is not bijective.
5. $$f(x) = -(x-3)^2+4;\ f^{-1}(y) = 3+\sqrt{(4-y)}$$
6. $$g^{-1}(y) = 1/y - 1$$

Self-Study 4.4: Maps

4.4b(i) · Form f∘e

1. Every real x is allowed: $e^x \in \mathbb{R}$, the domain of f.
2. $(f\circ e)(x) = -(e^x)^2 + 6e^x - 5 = -e^{2x} + 6e^x - 5$.
3. Writing $t = e^x > 0$ gives $4 - (t-3)^2$. Its maximum 4 occurs when $t=3$; as $t \to 0^+$ or $t \to \infty$ the value tends without bound downward. The range is $(-\infty, 4]$.
4. $$(f \circ e)(x) = -e^{2x} + 6e^x - 5; \text{ range } (-\infty, 4]$$

Self-Study 4.4: Maps

4.4b(ii) · Form g∘f

1. Require $4 - (x-3)^2 \geq 0$, so $(x-3)^2 \leq 4$ and $1 \leq x \leq 5$. This is the composite domain.
2. On it, f(x) ranges over $[0,4]$, and $(g\circ f)(x) = 1/(1+f(x)) = 1/[5-(x-3)^2]$.
3. The denominator runs from 1 to 5; reciprocation reverses the order. The range is $[1/5, 1]$.
4. $$(g \circ f)(x) = 1/[5 - (x-3)^2], \text{ domain } [1, 5], \text{ range } [1/5, 1]$$

Self-Study 4.4: Maps

4.4b(iii) · Form g∘e

1. For all $x \in \mathbb{R}$, $e^x > 0$, so the composition is defined everywhere.
2. $(g\circ e)(x) = 1/(1+e^x)$.
3. As e^x ranges over $(0,\infty)$, this expression decreases through $(0,1)$. It approaches 1 and 0 but reaches neither, so the range is $(0,1)$.
4. $$(g \circ e)(x) = 1/(1 + e^x), \text{ range } (0, 1)$$

Self-Study 4.4: Maps

4.4b(iv) · Form g∘g

1. The composite domain is $[0,\infty)$, since $g(x) \geq 0$ for every allowed x.
2. $(g\circ g)(x) = 1/[1+1/(1+x)] = (1+x)/(2+x)$.
3. This increases from $1/2$ at $x=0$ and approaches 1 without reaching it. The range is $[1/2, 1)$.
4. $$(g \circ g)(x) = (1 + x)/(2 + x), \text{ range } [1/2, 1)$$

Self-Study 4.5: Maps

4.5i · If g∘f is injective, show f is injective

1. Suppose $f(x_1)=f(x_2)$. Applying g gives $(g\circ f)(x_1)=(g\circ f)(x_2)$.
2. Injectivity of g∘f implies $x_1=x_2$. Thus f is injective.
3. $$g \circ f \text{ injective } \Rightarrow f \text{ injective}$$

Self-Study 4.5: Maps

4.5ii · If g∘f is surjective, show g is surjective

1. Let $z\in Z$. Since g∘f is onto Z, some $x\in X$ satisfies $g(f(x))=z$.
2. The element $y=f(x)$ belongs to Y and $g(y)=z$. Every z has a preimage under g, so g is surjective.
3. $$g \circ f \text{ surjective } \Rightarrow g \text{ surjective}$$

Self-Study 4.5: Maps

4.5iii · If f and g are surjective, show g∘f is surjective

1. Let $z\in Z$. Surjectivity of g gives $y\in Y$ with $g(y)=z$.
2. Surjectivity of f gives $x\in X$ with $f(x)=y$. Therefore $g(f(x))=z$.
3. Every z is reached by g∘f, proving surjectivity.
4. $$f \text{ and } g \text{ surjective } \Rightarrow g \circ f \text{ surjective}$$

Self-Study 4.5: Maps

4.5iv · Give an injective composite with non-injective g

1. Let $X=\{a\}$, $Y=\{0,1\}$, $Z=\{z\}$. Define $f(a)=0$ and define $g(0)=g(1)=z$.
2. The map g is not injective because $0\neq 1$ but $g(0)=g(1)$.
3. The composite $X\to Z$ sends the only input a to z. A map on a singleton is injective, so g∘f is injective.
4. $$X = \{a\},\ Y = \{0,1\},\ Z = \{z\};\ f(a) = 0,\ g(0) = g(1) = z$$

Self-Study 4.6: Elementary functions

4.6a · Find and factor the roots

1. For f, divide by 5 and set $u=x^2$: $u^2+4u-5=(u-1)(u+5)$. Thus $f(x)=5(x^2-1)(x^2+5)$.
2. The real roots are $x=\pm 1$. Over $\mathbb{C}$ the complete linear factorization is $5(x-1)(x+1)(x-i\sqrt{5})(x+i\sqrt{5})$.
3. For $g_a(x)=ax^2+2x+1$: if $a=0$ it is $2x+1$, with one root $-1/2$. If $a\neq 0$, its discriminant is $\Delta=4-4a$.
4. When $a<1$ and $a\neq 0$ there are two distinct real roots $r_\pm=(-1\pm\sqrt{(1-a)})/a$, so $g_a=a(x-r_+)(x-r_-)$. At $a=1$ it is $(x+1)^2$.
5. When $a>1$ there are no real roots; over $\mathbb{C}$ its factors are $a(x-r_+)(x-r_-)$, where $r_\pm=(-1\pm i\sqrt{(a-1)})/a$.
6. $$f(x) = 5(x-1)(x+1)(x-i\sqrt{5})(x+i\sqrt{5})$$
7. $$\Delta(g_a) = 4(1-a)$$

Self-Study 4.6: Elementary functions

4.6b · Show cos(3x) is not generally 3cos x

1. Take $x=0$: $\cos(3\cdot 0)=1$ while $3\cos 0=3$, so they are not equal in general.
2. Use $\cos(2x+x)=\cos(2x)\cos x-\sin(2x)\sin x$. Substitute $\cos 2x=\cos^2 x-\sin^2 x$ and $\sin 2x=2\sin x\cos x$.
3. This gives $\cos 3x=\cos^3 x-3\sin^2 x\cos x=\cos^3 x-3(1-\cos^2 x)\cos x=4\cos^3 x-3\cos x$.
4. $$\cos(3x) = 4\cos^3 x - 3\cos x$$

Self-Study 4.7: Strictly monotone functions

4.7a · Increasing implies injective

1. Take $x\neq y$. Either x
2. If y
3. $$x$$

Self-Study 4.7: Strictly monotone functions

4.7b · Decreasing implies injective

1. Take $x \neq y$. If $x < y$ then $f(x) > f(y)$; if $y < x$ then $f(y) > f(x)$.
2. In either case the outputs differ, so f is injective.
3. $$x < y \Rightarrow f(x) \neq f(y)$$

Self-Study 4.7: Strictly monotone functions

4.7c · Is every injection strictly monotone?

1. Define $f:\mathbb{R}\to\mathbb{R}$ by swapping 0 and 1 and leaving every other real number fixed. This is a bijection, hence injective.
2. Since $0<1/2$ but $f(0)=1>f(1/2)=1/2$, it is not increasing. Since $1/2<1$ but $f(1/2)=1/2>f(1)=0$, it is not decreasing.
3. Thus injectivity does not imply strict monotonicity.
4. $$f(0) = 1 > f(1/2) = 1/2,\ f(1/2) = 1/2 > f(1) = 0$$

Self-Study 4.7: Strictly monotone functions

4.7d · Conclude that exp has an inverse

1. The exponential is strictly increasing on $\mathbb{R}$, so by part (a) it is injective.
2. Its range is $(0,\infty)$, so $\exp:\mathbb{R}\to(0,\infty)$ is also surjective.
3. It is therefore bijective and has an inverse, the natural logarithm $\ln:(0,\infty)\to\mathbb{R}$.
4. $$\exp : \mathbb{R} \to (0,\infty) \text{ is bijective; } \exp^{-1} = \ln$$

Self-Study 4.8: Tangent addition theorems

4.8a · Prove the tangent addition and subtraction formulas

1. The addition identities give $\sin(x\pm y)=\sin x\cos y\pm\cos x\sin y$ and $\cos(x\pm y)=\cos x\cos y\mp\sin x\sin y$.
2. The hypotheses ensure $\cos x$, $\cos y$, and $\cos(x\pm y)$ are nonzero, so division is valid. Divide top and bottom by $\cos x\cos y$.
3. The result is $\tan(x\pm y)=(\tan x\pm\tan y)/(1\mp\tan x\tan y)$. For the plus case use matching upper/lower signs; for the minus case use opposite signs.
4. $$\tan(x \pm y) = (\tan x \pm \tan y)/(1 \mp \tan x \tan y)$$

Self-Study 4.8: Tangent addition theorems

4.8b · Derive the tangent double-angle formula

1. Use $\tan(x+x)=(\tan x+\tan x)/(1-\tan^2 x)$.
2. Thus $\tan(2x)=2\tan x/(1-\tan^2 x)$, whenever x and 2x avoid the excluded tangent poles.
3. $$\tan(2x) = 2\tan x/(1-\tan^2 x)$$

Self-Study 4.8: Tangent addition theorems

4.8c · Compute tan(π/12) exactly

1. Use $\tan(\pi/3)=\sqrt{3}$ and $\tan(\pi/4)=1$. Then $\tan(\pi/12)=(\sqrt{3}-1)/(1+\sqrt{3})$.
2. Multiply numerator and denominator by $\sqrt{3}-1$: the denominator becomes $3-1=2$ and the numerator becomes $(\sqrt{3}-1)^2=4-2\sqrt{3}$.
3. Therefore $\tan(\pi/12)=2-\sqrt{3}$, a positive value as expected for an angle of 15°.
4. $$\tan(\pi/12) = 2-\sqrt{3}$$

Challenge 4.A: Surjective maps

4.A setup · Interpret s(k) and the target

1. For a fixed k-element set A, s(k) is the number of maps $f:M\to A$ whose image is all of A.
2. A map $M\to N$ is surjective exactly when its image has n elements, so s(n) counts the onto maps from M to N.
3. Assume $m=|M|\geq 1$ for the stated Step 1 recurrence. If $m=0$, there is one onto map when $n=0$ and none when $n>0$; that case is separate from this recurrence.
4. $$f \text{ is onto} \Leftrightarrow |f(M)| = |N| = n$$

Challenge 4.A: Surjective maps

4.A Step 1 · Derive the recurrence for s(k)

1. There are k^m total maps M→A. If the image has size j, first choose its j-element image B in C(k,j) ways, then choose a map onto B in s(j) ways.
2. Summing over image sizes gives $k^m=\Sigma_{j=1}{}^{k}\, C(k,j)s(j)$. There is no empty image because $m\geq 1$.
3. The j=k term is s(k), since C(k,k)=1. Rearranging gives $s(k)=k^m-\Sigma_{j=1}{}^{k-1}\, C(k,j)s(j)$.
4. $$k^m = \sum_{j=1}^{k} C(k,j)s(j);\ s(k) = k^m - \sum_{j=1}^{k-1} C(k,j)s(j)$$

Challenge 4.A: Surjective maps

4.A Step 2 · Check the proposed formula for k=1, 2, and 3

1. For k=1, $s(1)=1^m=1$. The proposed sum is $C(1,1)1^m=1$.
2. For k=2, $s(2)=2^m-2s(1)=2^m-2$. The guess gives $C(2,1)(-1)^1 1^m+C(2,2)2^m=2^m-2$.
3. For k=3, $s(3)=3^m-3s(1)-3s(2)=3^m-3\cdot 2^m+3$. The guess gives the same alternating expression. These cases support, but do not prove, $s(k)=\Sigma_{j=1}{}^{k}\, C(k,j)(-1)^{k-j}j^m$.
4. $$s(1) = 1;\ s(2) = 2^m - 2;\ s(3) = 3^m - 3\cdot 2^m + 3$$

Challenge 4.A: Surjective maps

4.A Step 3a · Reverse the order of a triangular sum

1. The left-hand iterated sum first chooses j=1,…,k, then i=1,…,j. Its index pairs are exactly $\{(i,j):1\leq i\leq j\leq k\}$.
2. If instead i is fixed first, i ranges from 1 to k and j ranges from i to k. Thus $\Sigma_{j=1}{}^{k}\Sigma_{i=1}{}^{j}(\ldots) = \Sigma_{i=1}{}^{k}\Sigma_{j=i}{}^{k}(\ldots)$.
3. Both sides add the same term once for every pair in that finite triangular region.
4. $$\sum_{j=1}^{k}\sum_{i=1}^{j}(\ldots) = \sum_{i=1}^{k}\sum_{j=i}^{k}(\ldots)$$

Challenge 4.A: Surjective maps

4.A Step 3b · Prove the binomial coefficient identity

1. For $0\le i\le j\le k$, $C(k,j)C(j,i)=k!/[j!(k-j)!]\cdot j!/[i!(j-i)!]$.
2. Cancel $j!$ to get $k!/[i!(j-i)!(k-j)!]$.
3. On the other hand, $C(k,i)C(k-i,j-i)=k!/[i!(k-i)!]\cdot(k-i)!/[(j-i)!(k-j)!]$, the same expression.
4. $$C(k,j)C(j,i) = C(k,i)C(k-i,j-i)$$

Challenge 4.A: Surjective maps

4.A Step 3c · Establish the convolution identity

1. Start with $\Sigma_{j=1}^{k} C(k,j)g(j)$ and insert $g(j)=\Sigma_{i=1}^{j} C(j,i)(-1)^{j-i} i^m$.
2. Use Step 3a to reverse the order, then Step 3b: $C(k,j)C(j,i)=C(k,i)C(k-i,j-i)$.
3. The result is $\Sigma_{i=1}^{k} C(k,i)i^m \Sigma_{j=i}^{k} C(k-i,j-i)(-1)^{j-i}$. Put $r=j-i$. The inner sum is $\Sigma_{r=0}^{k-i} C(k-i,r)(-1)^r=(1-1)^{k-i}$.
4. It is zero unless $i=k$, when it equals 1. The whole expression therefore reduces to k^m, proving $k^m=\Sigma_{j=1}^{k} C(k,j)g(j)$.
5. $$k^m = \sum_{j=1}^{k} C(k,j)g(j)$$

Challenge 4.A: Surjective maps

4.A Step 4 · Prove s(k)=g(k) by induction

1. For $k=1$, $s(1)=1$ and $g(1)=C(1,1)(-1)^0 1^m=1$.
2. Assume $s(j)=g(j)$ for every $1\le j<k$. Then $s(k)=k^m-\Sigma_{j=1}^{k-1} C(k,j)s(j)$.
3. The convolution identity says $k^m=\Sigma_{j=1}^{k} C(k,j)g(j)$, so isolating its final term gives $g(k)=k^m-\Sigma_{j=1}^{k-1} C(k,j)g(j)$.
4. The induction hypothesis makes these right sides equal, hence $s(k)=g(k)$. This proves the formula for every $k\ge 1$.
5. $$s(k) = g(k) = \sum_{j=1}^{k} C(k,j)(-1)^{k-j}j^m$$

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